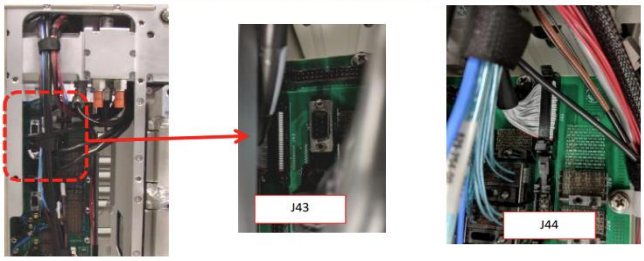
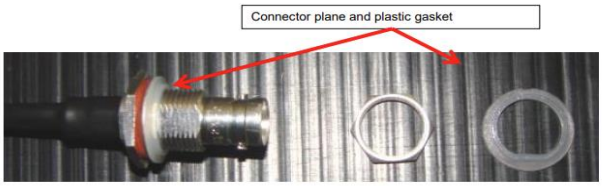
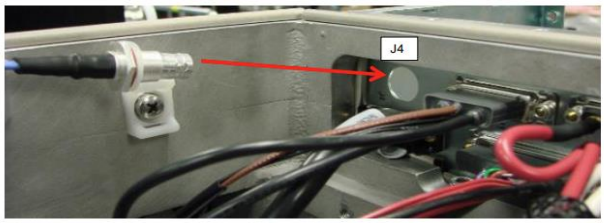
Build Overview:

1 Install the following cables:

- **a.SB Sync**
- **b.CLK Fanout**
- **c.New cable**
- **d.TDR Coax**
- **e.CLK100 Loopback**
- **f.CD Cal Bus**
- **g.HV Interlock**
- **h.DRG Support**







CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

MAN:

CERTIFIED TECHNICIANS.

MATERIAL:

NO.	PART NO.	DESCRIPTION	QTY.
1	650-695-00	CABLE CABLE 12SLOT DRGSPT TO DBMISC 26 TELECOM WITH CONN	1
2	650-695-01	CABLE CABLE 12SLOT DRGSPT TO DBMISC 26 TELECOM WITH CONN	1
3	627-718-02	CABLE DRG SBDB2 SYNCX TELECOM WITH CONN	2
4	813-929-01	HDW VELCRO STRP 5/8	4
5	629-254-00	CABLE UF2_CLKFANOUT12 LCXCEDE TELECOM WITH CONN	1
6	627-083-00	CABLE UF2 TDR COAX TELECOM WITH CONN	2
7	650-679-00	CABLE UF2 12SLOT TH_INT_SMCLON_PRESS TELECOM WITH CONN	1
8	659-442-02	CABLE UF2 CLK100 TH BULKHEAD W SOLDER TELECOM WITH CONN	1
9	650-698-00	CABLE UF2 12SLOT DC_CAL_BUS TELECOM WITH CONN	1
10	650-698-01	CABLE UF2 12SLOT DC_CAL_BUS TELECOM WITH CONN	1
11	636-665-00	CABLE UF2_NVRAM_COMM 12 V W/ CONN	1
12	655-265-01	CABLE CABLE_HV_INTLK_INT TALL CONN TELECOM WITH CONN	1

METHOD:

Flow chart of operation steps

BPM preparation

Preparation Tools

Prepare Materials

Complete Operation

Inventory Materials

Operation Check

Inventory tool

TOOL NAME	TORQUE VALUE
Flat-blade screwdriver	0.19N.m
3/16 sleeve torque	0.19N.m
14mm socket	0.7N.m.
open-end wrench	N/A

MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
1	650-695-00	CABLE CABLE 12SLOT DRGSPT TO DBMISC 26 TELECOM WITH CONN	1
2	650-695-01	CABLE CABLE 12SLOT DRGSPT TO DBMISC 26 TELECOM WITH CONN	1

MACHINE:

N/A

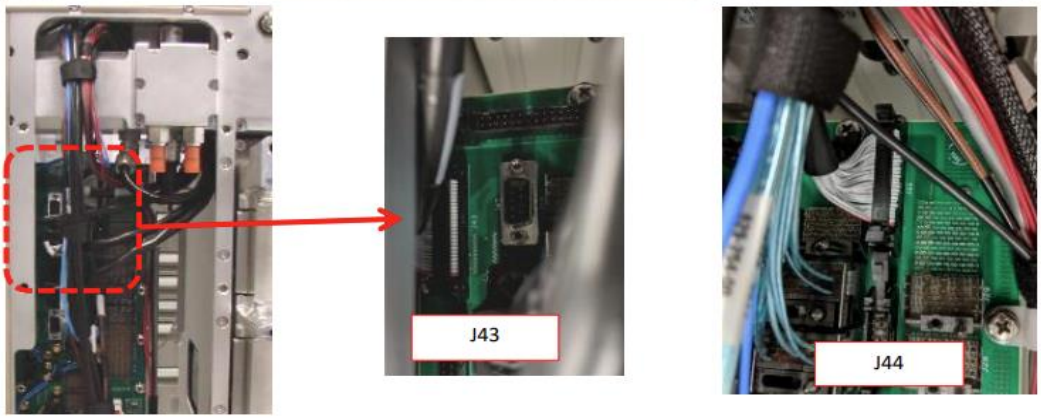
METHOD:

Operation steps

1.1 Take one DRGSPTX cable (650-695-00) and one cable (650-695-01).

1.2 Pass the cables marked SUPPLN_44 and SUPPLN_43 through the opening of the T/H, and through the cable support in the T/H, out of the backplane side.

1.3 Insert the wires into the J44 and J43 connectors on the back panel as shown in the figure, and tie the wire harness away from the DB side with existing cable tie.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
3	627-718-02	CABLE DRG SBDB2 SYNCX TELECOM WITH CONN	2

MACHINE:

Marker

METHOD:

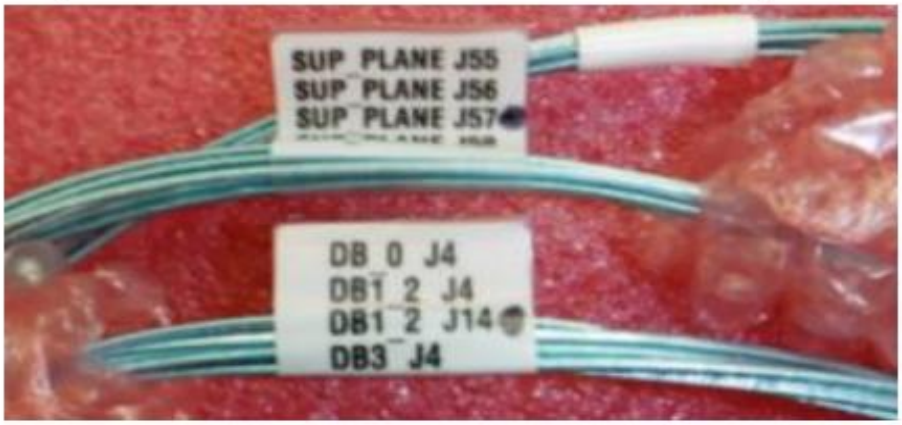
2.1 Install 2 DRG SBDB2 syncx cables (627-718-02).

2.2 Label the lines in the following order:

a. Mark "SUP PLANE J57" on the backplane side and "DB1_2J14" on the connector side.

b. Mark "SUP PLANE J56" on the backplane side and "DB1_2J4" on the connector side.

Mark the labels "SUP PLANE J57" and DB1_2J14".
Repeat the above label on other lines.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

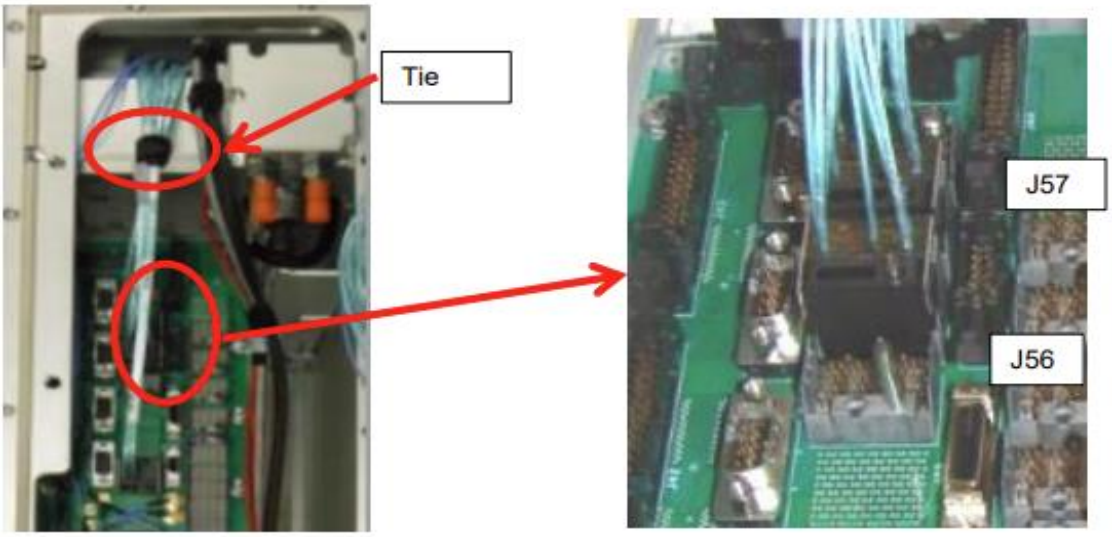
ITEM	PART NUMBER	DESCRIPTION	QTY
4	813-929-01	HDW VELCRO STRP 5/8	1

MACHINE:

N/A

METHOD:

2.3 Insert the wires corresponding to J57 and J56 on the back panel, and tie them with a cable tie 813-929-01.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
5	813-929-01	HDW VELCRO STRP 5/8	1

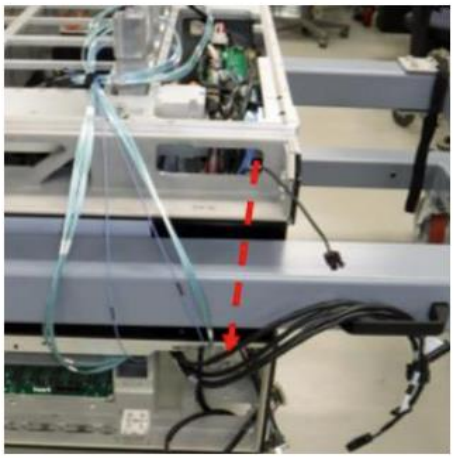
MACHINE:

N/A

METHOD:

2.4 Pass the cables from the T/H backplane area through the openings shown below, and then route them to the DB12 side through the channel.

2.5 Wrap the extra cable on the IPD post, and do not bend or twist the cable. Use one cable tie 813-929-01 to fix it.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

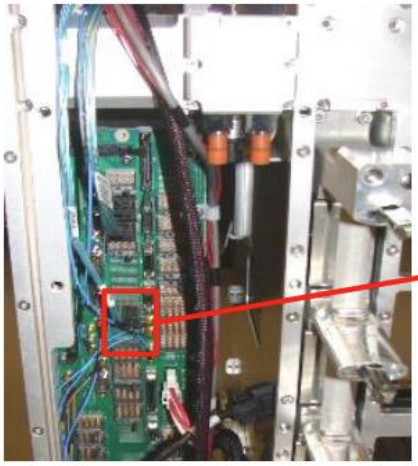
ITEM	PART NUMBER	DESCRIPTION	QTY
6	629-254-00	CABLE UF2_CLKFANOUT12 LCXCEDE TELECOM WITH CONN	1

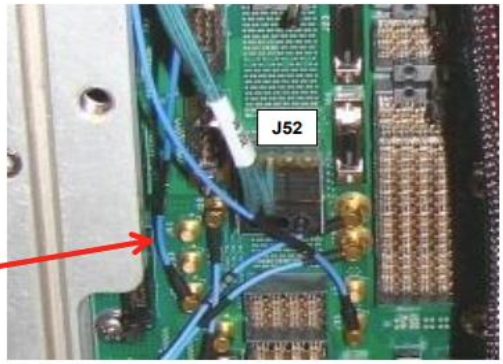
MACHINE:

N/A

METHOD:

3.1 Install one CLK FANOUT line (629-254-00).
3.2 Insert the connector labeled "J52" on the line into the backplane end.
3.3 Pass the cables over the manifold and then route them to the DB side through the channel,
Wrap the extra cable on the IPD post, do not bend or twist the cable, use the same cable tie to tie the cable tightly.





MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
7	627-083-00	CABLE UF2 TDR COAX TELECOM WITH CONN	2

MACHINE:

N/A

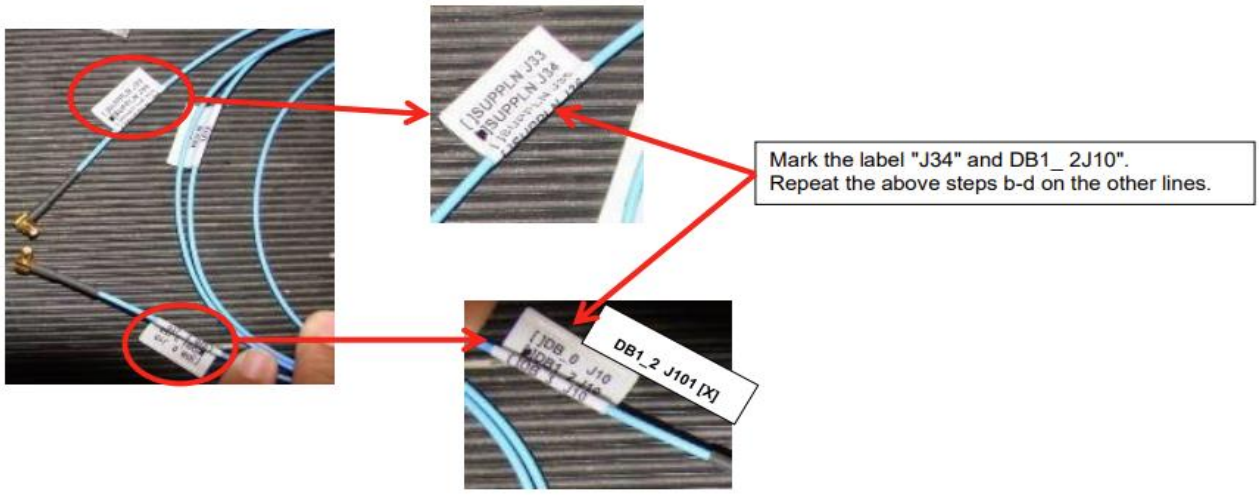
METHOD:

4.1 Install 2 TDR COAX cables (627-083-00).

4.2 Label the lines in the following order:

a. Mark "SUPPLN J34" on the back panel end, and mark "DB1_2 J10" on the other end.

b. Mark "SUPPLN J35" on the back panel end, and mark "DB1_2 J11" on the other end.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

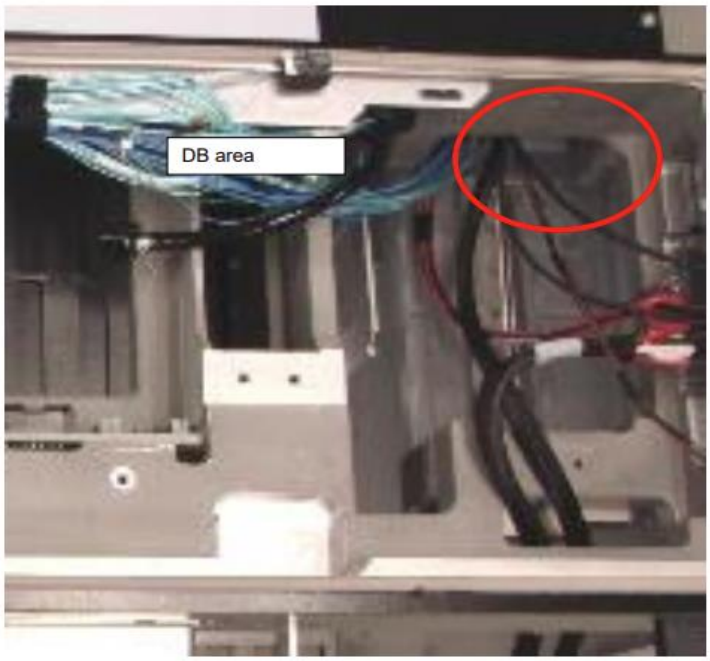
N/A

MACHINE:

N/A

METHOD:

4.3 Pass the cables from the T/H backplane area through the openings shown below,
and then route them through the channel to the DB12 side.
Wrap the extra cable on the IPD post, do not bend or twist the cable, use the same cable tie to tie the cable tightly.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

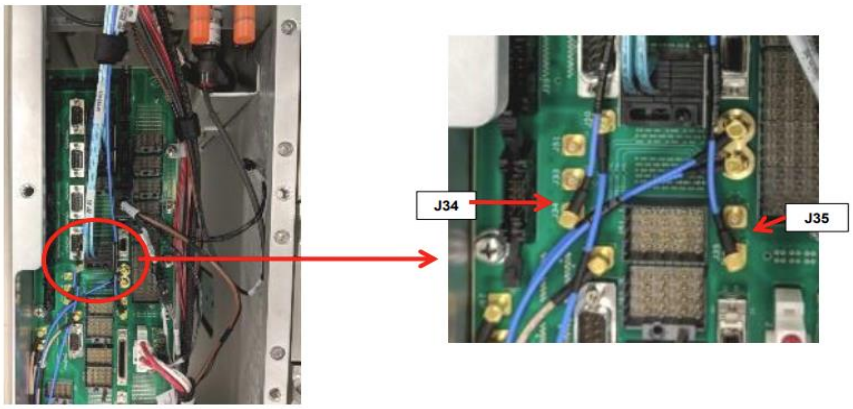
MACHINE:

N/A

METHOD:

4.4 Route the cable along the edge of the T/H frame and insert it into the back panel ends J35 and J34 according to the label.

4.5 Wrap the extra cable on the IPD column, do not bend or twist the cable, use the same cable tie to tie the cable tightly.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
8	650-679-00	CABLE UF2 12SLOT TH_INT_SMCLON_PRESS TELECOM WITH CONN	1

MACHINE:

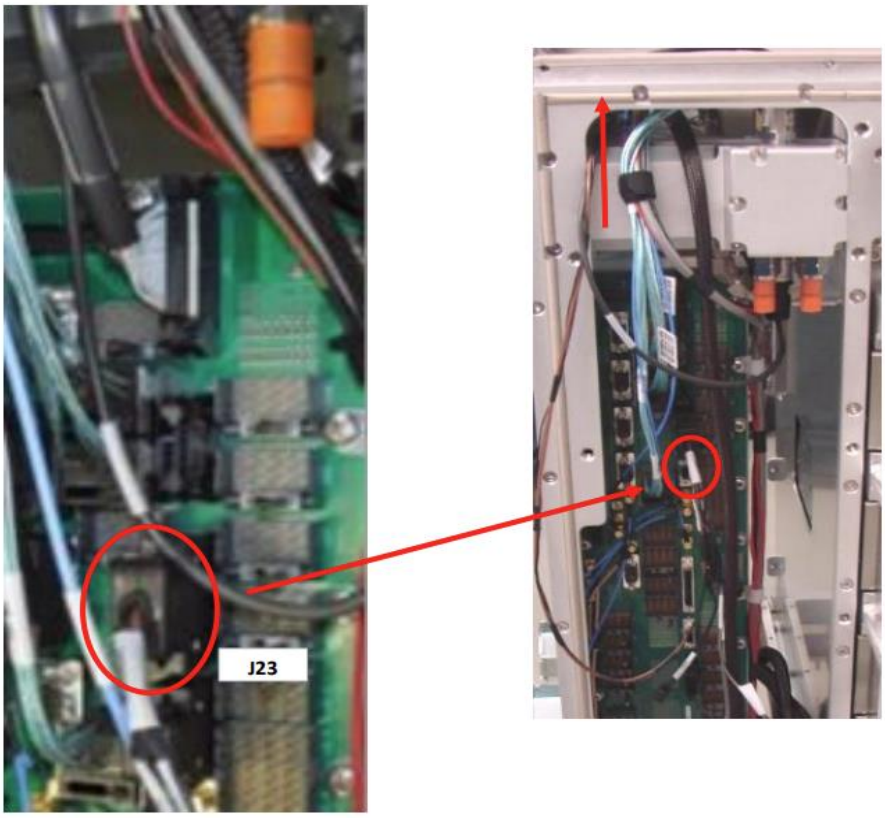
N/A

METHOD:

5.1 Install one CLK100 LOOPBACK cable (650-679-00), and connect the large black connector as labeled to the backplane "J23".

5.2 As shown in the figure, place the J1 interface of the cable through the T/H frame channel on the opposite panel. Install the J1 interface later.

5.3 Separate the two pressure sensor wires. The long wire is placed at the back plate end (Supply), and the short wire is placed at the outlet pressure sensor (return) as shown.



Safety Tips: N/A

MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

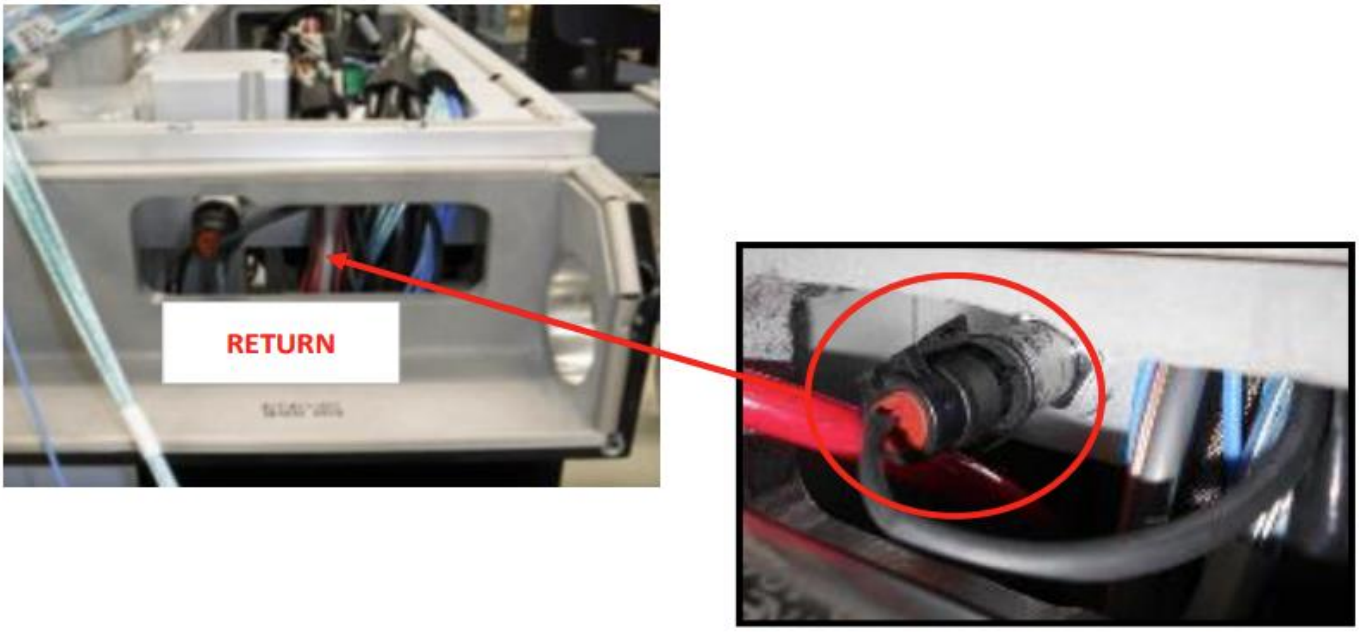
N/A

MACHINE:

N/A

METHOD:

5.4 Install the long pressure sensor connector to the Manifold pressure sensor as shown.
Insert the connector into the RETURN pressure sensor according to the position of the buckle.
After the card is locked, confirm whether it is in place.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

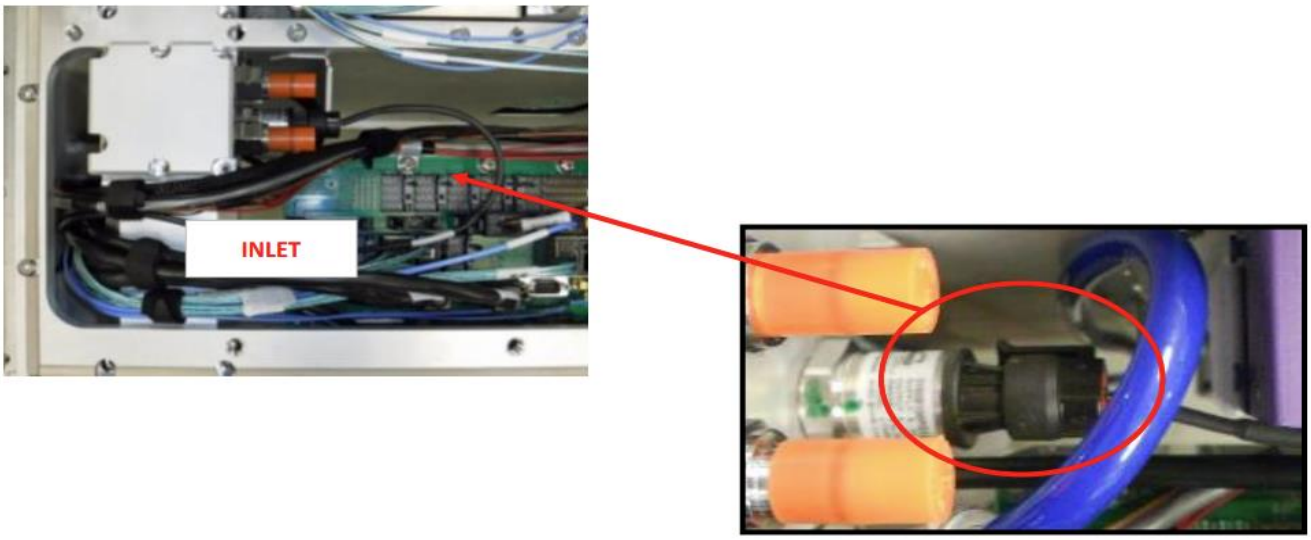
N/A

MACHINE:

N/A

METHOD:

5.5 Install the shorter pressure sensor connector to the Manifold pressure sensor as shown
Insert the connector into the INLET pressure sensor according to the position of the buckle.
After the card is locked, confirm whether it is in place.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

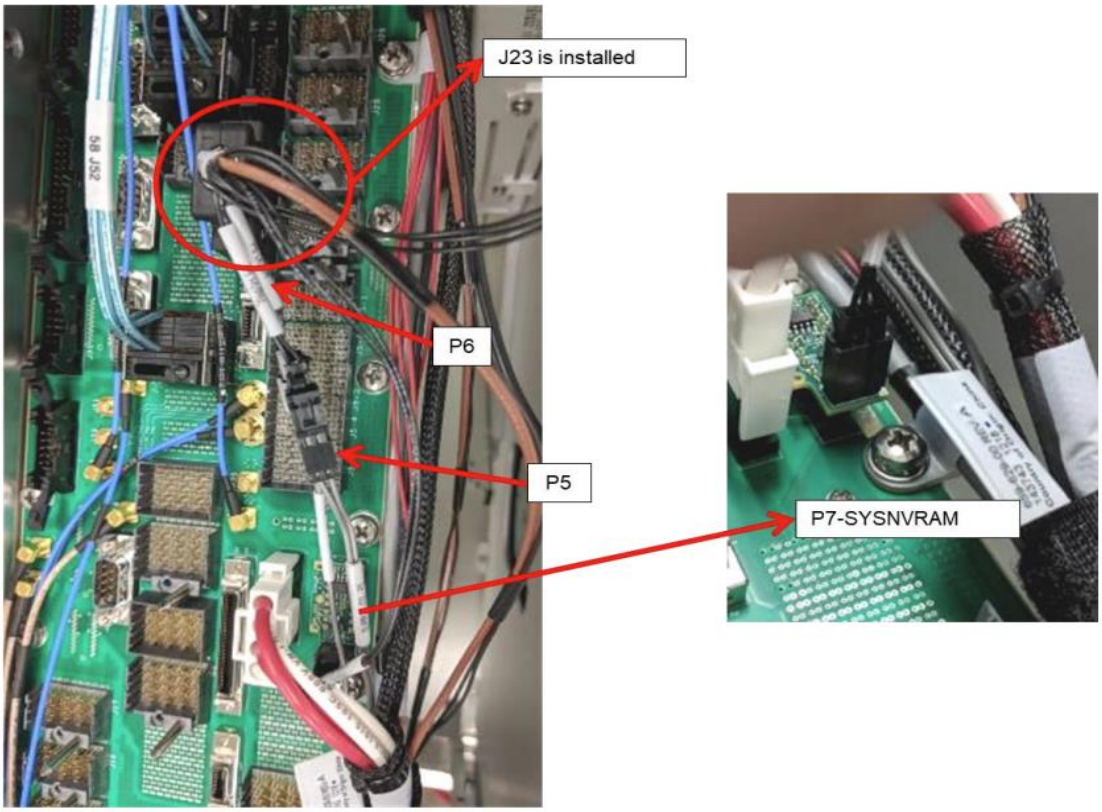
N/A

METHOD:

5.6 After installing the J23 line, insert the P5, P6 and P7 connectors into the key switch, door lock switch and PCBA board SYSNVRAM.

5.7 "P8 AIR2" connector is empty and need not be connected.

5.8 Route the cable according to the existing harness, and use the existing cable ties to tighten the cable tightly.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
9	659-442-02	CABLE UF2 CLK100 TH BULKHEAD W SOLDER TELECOM WITH CONN	1

MACHINE:

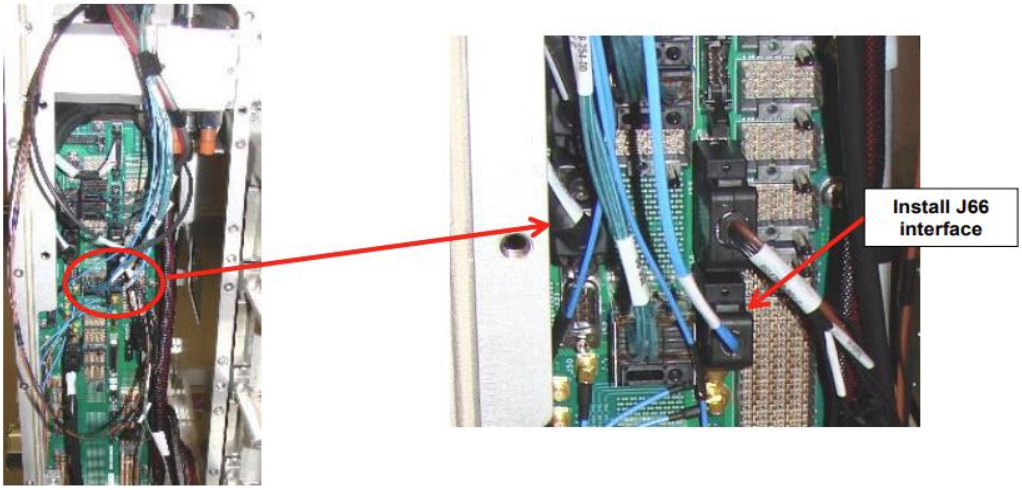
N/A

METHOD:

6.1 Install the CLK100 TH Bulkhead line (659-442-02).

6.2 Install the black plastic connector to the “J66” end of the backplane according to the on line label.

6.3 Place the J4 connector of the cable through the T/H frame channel on the opposite panel as shown in the figure. Install the J4 interface later.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
10	650-698-00	CABLE UF2 12SLOT DC_CAL_BUS TELECOM WITH CONN	1
11	650-698-01	CABLE UF2 12SLOT DC_CAL_BUS TELECOM WITH CONN	1

MACHINE:

Flat-blade screwdriver 0.19 N·m

METHOD:

6.4 Install the DC_CAL_BUS, 75V line (650-698-00, 650-698-01).
6.5 Insert the thread into the T/H open slot as shown in the figure,
and place it on the back plate end through the rack inside the T/H.
6.6 Insert the two wires according to the labels J30 and J46 as shown in the figure,
and fix them with a flat-blade screwdriver. The torque value is 0.19 N·m.
6.7 Place the two black wires as shown in the figure and bind the wire harness with the existing cable ties.

MAN:

CERTIFIED OPERATOR, TECHNICIAN

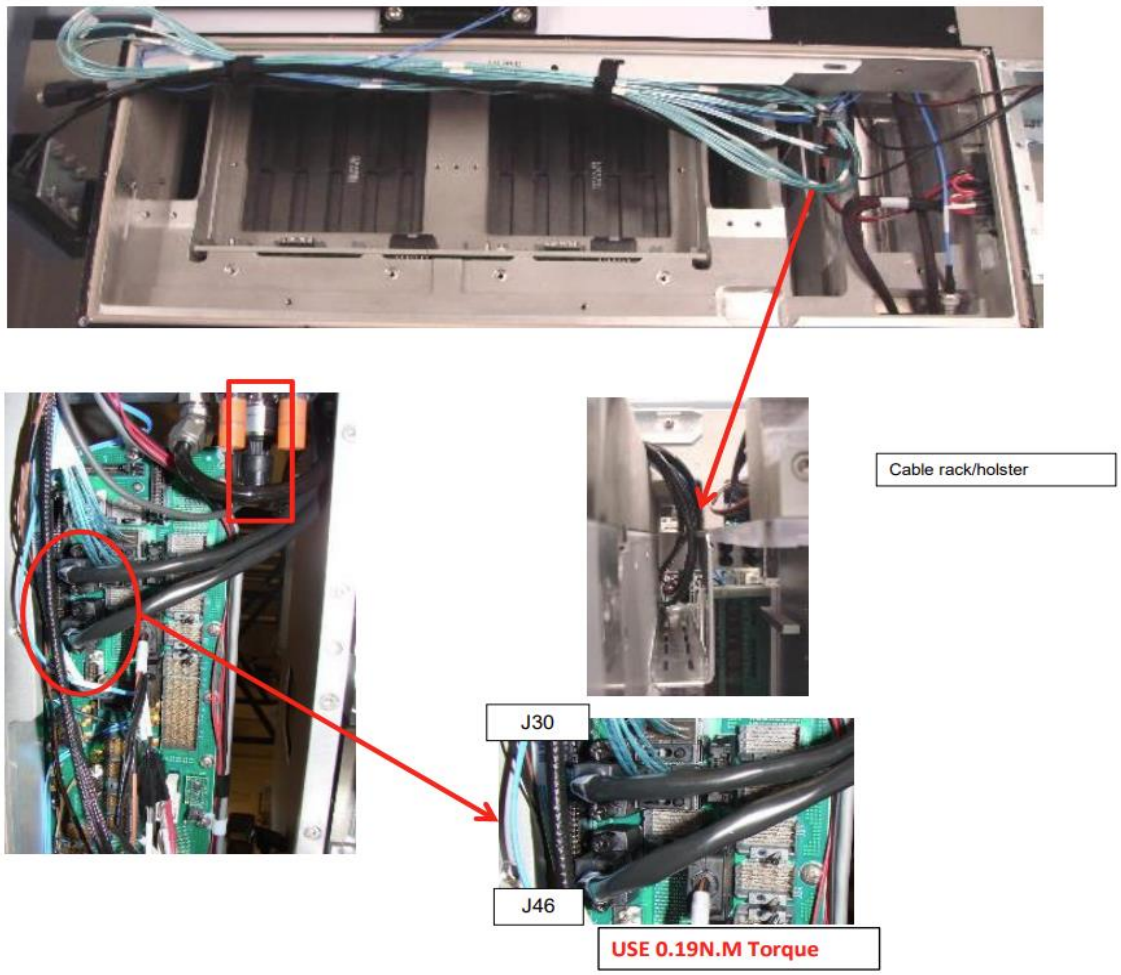
MATERIAL:

N/A

MACHINE:

N/A

METHOD:



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

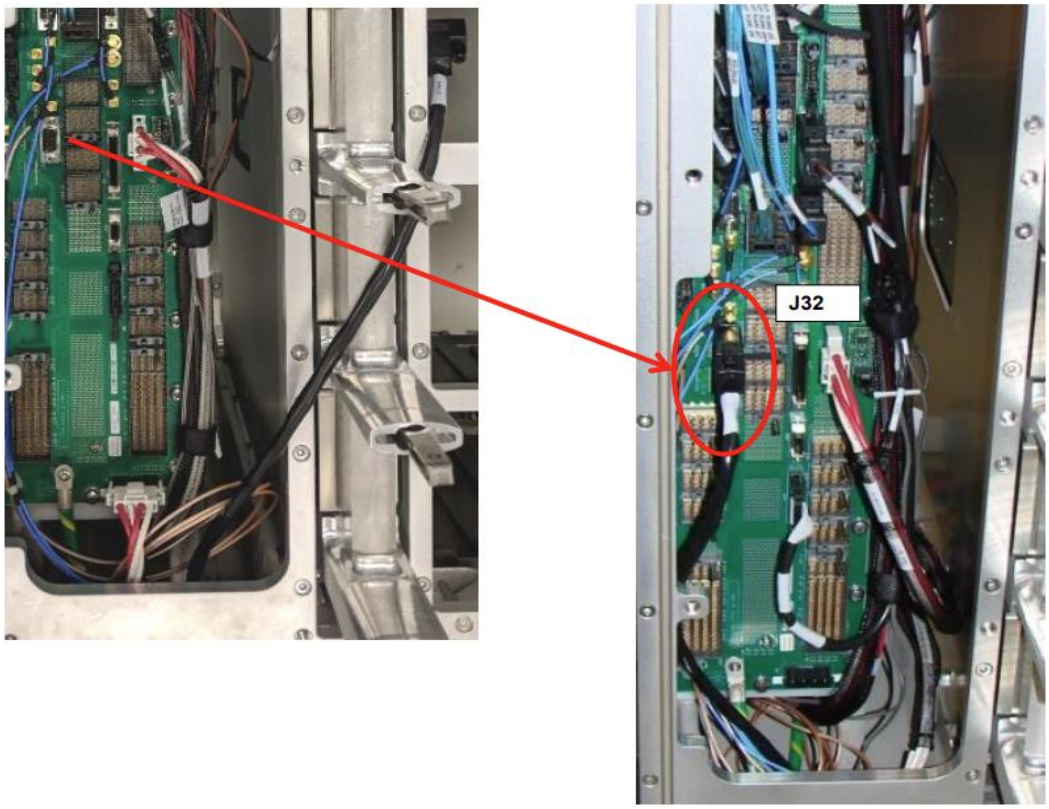
MACHINE:

Flat-blade screwdriver 0.19 N·m

METHOD:

6.8 Insert the previously installed Interlock cable 652-811-00 to the J32 interface of the backplane as shown in the figure

Use a flat-blade screwdriver to fix the torque value is 0.19 N·m.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
12	813-929-01	HDW VELCRO STRP 5/8	2

MACHINE:

N/A

METHOD:

6.9 Use two cable ties 813-929-01 to fix the wires as shown in the figure.

MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

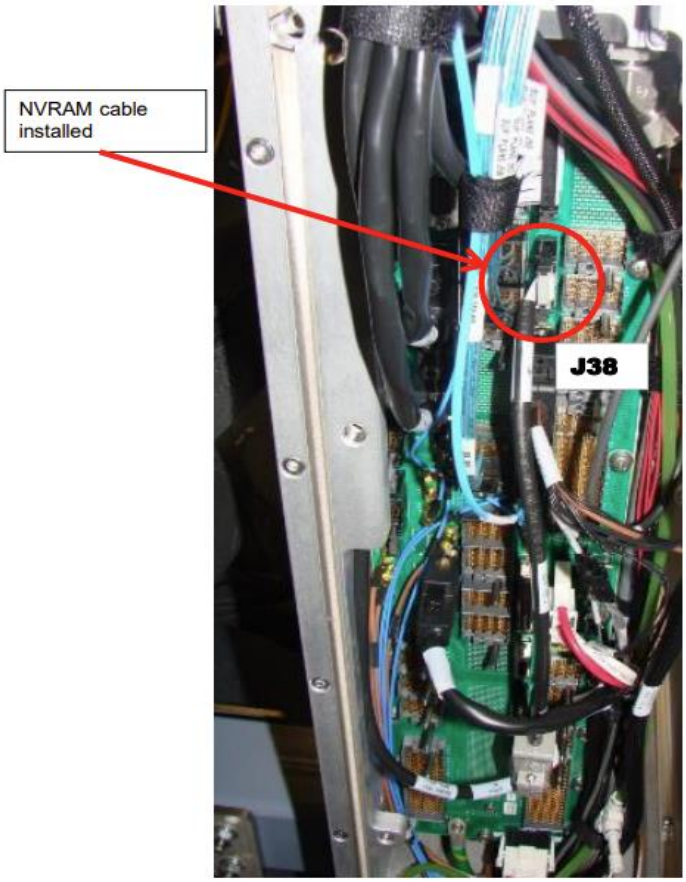
ITEM	PART NUMBER	DESCRIPTION	QTY
13	636-665-00	CABLE UF2_NVRAM_COMM 12 V W/ CONN	1

MACHINE:

N/A

METHOD:

6.10 Take one NVRAM cable 636-665-00 and connect it to the "J38" end of the backplane according to the line labeled "Support Plane J38".



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

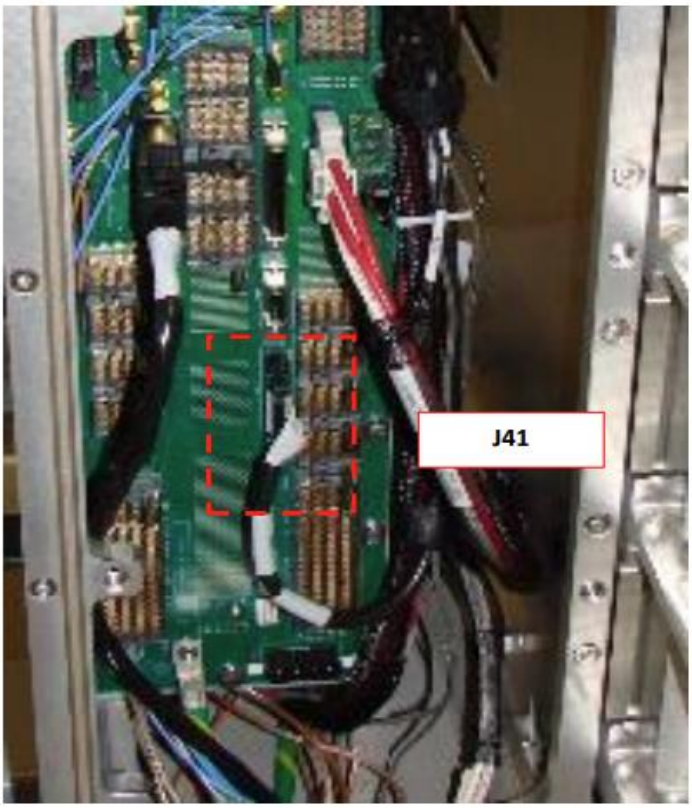
N/A

METHOD:

7.1 Take one HV_INTLK_INTERNAL cable 655-265-01, insert the cable into the backplane "J41" according to the label, and tie the cable and the EMO cable together.

7.2 Pass the thread from the opening next to the QD to the DB area.

7.3 Turn the T/H to the DB side upward position as shown in the figure.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

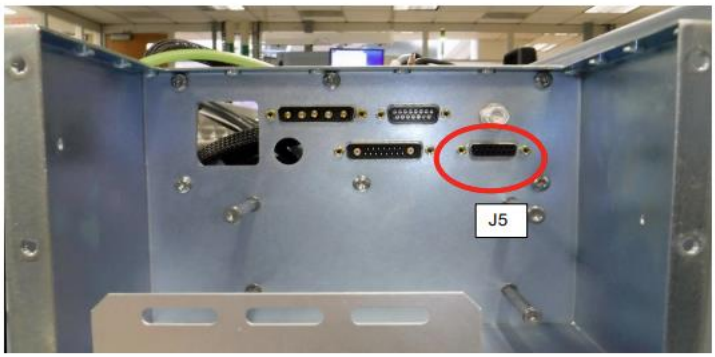
ITEM	PART NUMBER	DESCRIPTION	QTY
14	655-265-01	CABLE CABLE_HV_INTLK_INT TALL CONN TELECOM WITH CONN	1

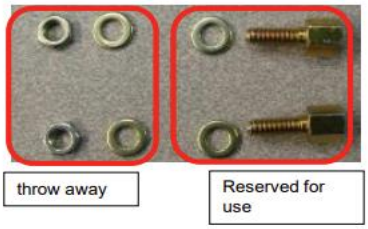
MACHINE:

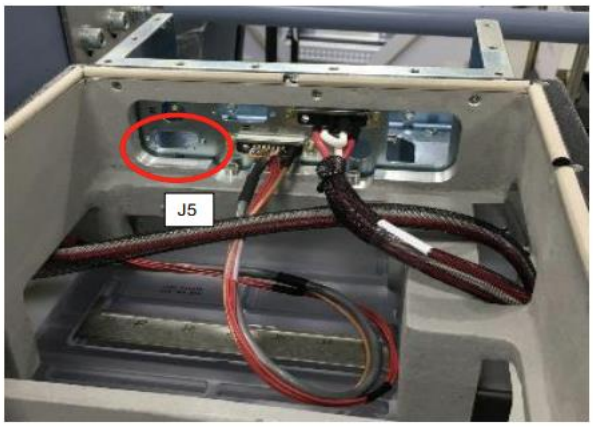
3/16 sleeve torque 0.19 N•m

METHOD:

7.4 Remove the screw washer from the connector "J5". Leave two for each flat washer and stud.
Throw away the extra gasket.
7.5 Connect J5 to the panel, use two flat washers and studs to fix the connector,
The tool is a 3/16 sleeve with a torque value of 0.19 N•m.







MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

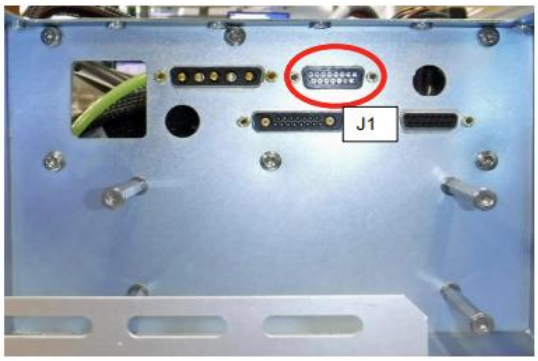
N/A

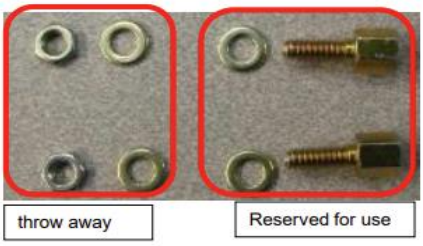
MACHINE:

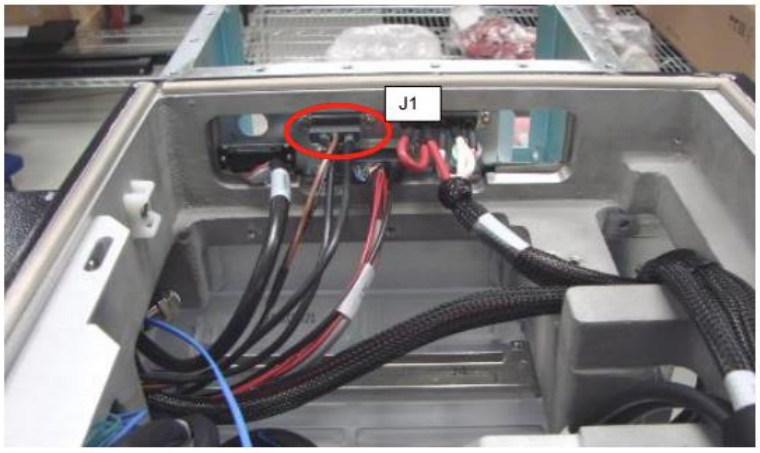
3/16 sleeve torque 0.19 N•m.

METHOD:

7.6 Remove the screw washer from the previously installed 650-679-00 connector "J1".
Leave two for each flat washer and stud. Throw away the extra gasket.
7.7 Connect J1 to the panel, use two flat washers and studs to fix the connector from inside the panel,
The tool is a 3/16 sleeve with a torque value of 0.19 N•m.







MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

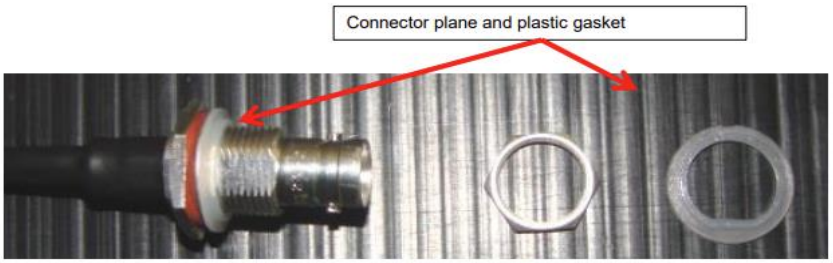
N/A

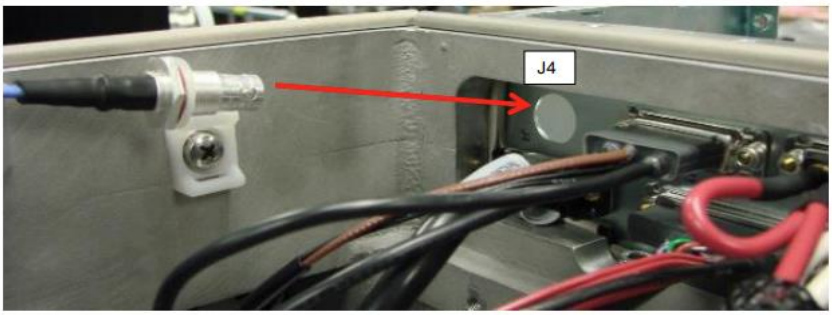
METHOD:

8.1 Remove the nut on the J4 connector of the 659-442-02 line and an adjacent plastic washer, save it and use it for subsequent assembly.

8.2 Install connector J4 on the panel, and ensure that the flat surface on the connector head is placed on the flat surface on the panel when installing.

When rotating the wire, make sure not to drop the gasket on the connector.





MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

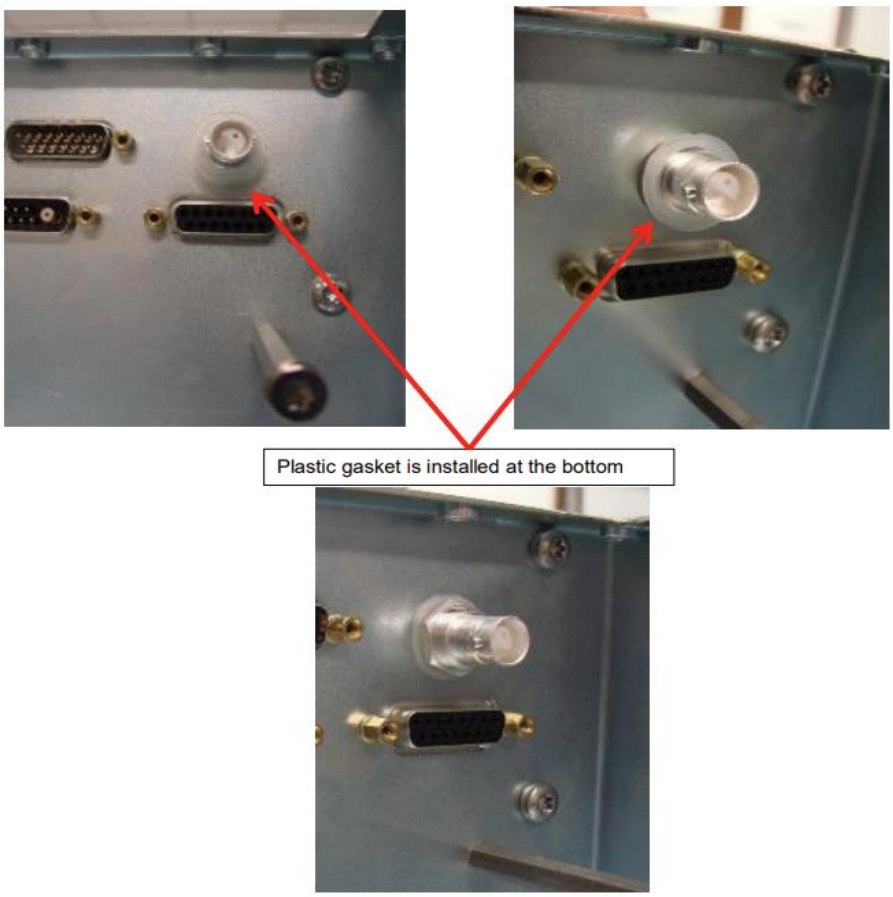
N/A

MACHINE:

14mm socket 0.7 N·m, open-end

METHOD:

8.3. Hex nuts and washers removed before installation inside the panel. Pay attention to the direction when installing the gasket to ensure that the plane of the gasket is consistent with the plane of the panel. Use a tool 14mm socket and open-end wrench to fix it, with a torque value of 0.7 N·m.



Safety Tips: N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

Matters needing attention

1. Do ESD and protective safety measures.
2. Be careful during the assembly process so as not to scratch the surface of the parts.
3. After assembly, check the assembled parts for defects and leak-free operation.